EXPERIMENT NO: 5

ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

AIM:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus.

ECA14 Embedded Systems Lab Manu…

SOFTWARE REQUIRED:

- Proteus 8 software
- Arduino IDE

PROGRAM :

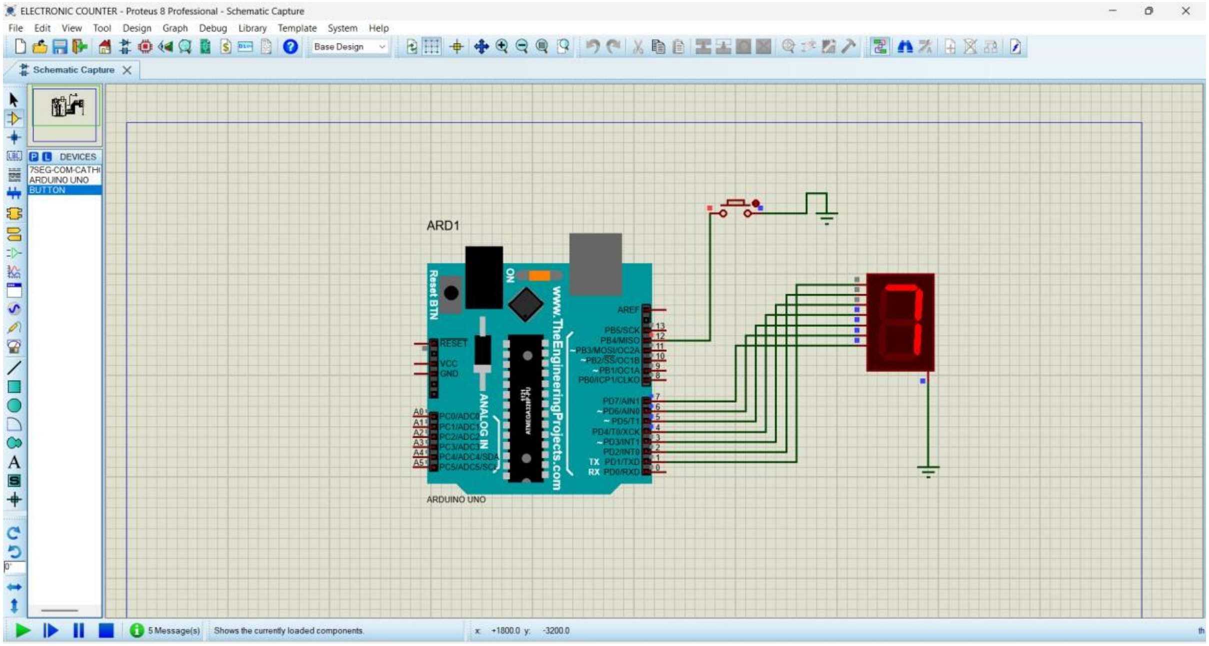
```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;

int delay_time=200;

void setup() {

 //configure pin2 as an input and enable the internal pull-up resistor

 pinMode(12, INPUT_PULLUP);

 pinMode(1,OUTPUT);  pinMode(2,OUTPUT);  pinMode(3,OUTPUT);
pinMode(4,OUTPUT);  pinMode(5,OUTPUT); pinMode(6,OUTPUT);
pinMode(7,OUTPUT);}

void loop() {

 int sensorVal = digitalRead(12);

 if (sensorVal == LOW) {

   x0=true;  }

 while(x0){

   zero();

   sensorVal = digitalRead(12);

   if (sensorVal == LOW) {

     x1=true;    x0=false;   }  }
```

```
while(x1){
  one();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x2=true;      x1=false;   }  }
while(x2){
  two();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x3=true;      x2=false;    }  }
while(x3){
  three();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x4=true;      x3=false;   }  }
while(x4){
  four();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x5=true;      x4=false;    }  }

while(x5){
  five();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x6=true;      x5=false;    }  }
while(x6){
```

```
    six();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
      x7=true;      x6=false;    }  }
  while(x7){
    seven();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
      x8=true;      x7=false;    }  }
  while(x8){
    eight();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
      x9=true;      x8=false;    }  }
  while(x9){
    nine();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
      x0=true;      x9=false;    }  }}
void zero(){
  for(int i=1;i<7;i++)  {
    digitalWrite(i,HIGH);    digitalWrite(7,LOW);  }
  delay(delay_time);}
void one(){
  digitalWrite(1,LOW);  digitalWrite(2,HIGH);  digitalWrite(3,HIGH);
digitalWrite(4,LOW);  digitalWrite(5,LOW);  digitalWrite(6,LOW);
digitalWrite(7,LOW);  delay(delay_time);}
void two() {
```

```
  digitalWrite(1,HIGH);  digitalWrite(2,HIGH);  digitalWrite(3,LOW);
digitalWrite(4,HIGH);  digitalWrite(5,HIGH);  digitalWrite(6,LOW);
digitalWrite(7,HIGH);  delay(delay_time);}

void three(){

digitalWrite(1,HIGH);  digitalWrite(2,HIGH);  digitalWrite(3,HIGH);
digitalWrite(4,HIGH);    digitalWrite(5,LOW);  digitalWrite(6,LOW);
digitalWrite(7,HIGH);  delay(delay_time);}

void four(){

  digitalWrite(1,LOW);  digitalWrite(2,HIGH);  digitalWrite(3,HIGH);
digitalWrite(4,LOW);  digitalWrite(5,LOW);  digitalWrite(6,HIGH);
digitalWrite(7,HIGH);  delay(delay_time);}

void five(){

  digitalWrite(1,HIGH);  digitalWrite(2,LOW);  digitalWrite(3,HIGH);
digitalWrite(4,HIGH);  digitalWrite(5,LOW); digitalWrite(6,HIGH);
digitalWrite(7,HIGH);  delay(delay_time);}

void six(){

  digitalWrite(1,HIGH);digitalWrite(2,LOW); digitalWrite(3,HIGH);
digitalWrite(4,HIGH);  digitalWrite(5,HIGH);  digitalWrite(6,HIGH);
digitalWrite(7,HIGH);  delay(delay_time);}

void seven(){

  digitalWrite(1,HIGH);  digitalWrite(2,HIGH);  digitalWrite(3,HIGH);
digitalWrite(4,LOW);  digitalWrite(5,LOW);  digitalWrite(6,LOW);
digitalWrite(7,LOW);  delay(delay_time);}

void eight(){

  digitalWrite(1,HIGH);  digitalWrite(2,HIGH);  digitalWrite(3,HIGH);
digitalWrite(4,HIGH);  digitalWrite(5,HIGH);  digitalWrite(6,HIGH);
digitalWrite(7,HIGH);  delay(delay_time);}

void nine(){

  digitalWrite(1,HIGH);  digitalWrite(2,HIGH);  digitalWrite(3,HIGH);
digitalWrite(4,HIGH);  digitalWrite(5,LOW);  digitalWrite(6,HIGH);
digitalWrite(7,HIGH);  delay(delay_time);}
```

OUTPUT :



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